MY PENS

Personal health tracker — master project overview

Updated March 2026 · Phases 1–4 complete · Cowork enhancements applied · Multi-critic synthesis

What is MY PENS?

MY PENS is the internal project name and developer acronym for five tracking modules: Measurements, Performance (training), Eating, Nutrition (weight/body composition), Sleep. It is a local-first personal health tracker running on Next.js 14, TypeScript, Tailwind, Prisma, and SQLite. No accounts, no cloud, no subscriptions — all data lives in a single SQLite file on your machine at localhost:3000.

The name stays internal. The public-facing product will be positioned and branded around its core promise — not the module architecture.

Core promise (all messaging leads here):

Know what you actually weigh.

Private weight tracking that separates real body change from travel, restaurant meals, carbs, alcohol, and hard training.

Tech stack

Layer	Technology
Framework	Next.js 14 (App Router)
Language	TypeScript
Styling	Tailwind CSS
Database	Prisma ORM + SQLite (local file)
Charts	Recharts
Icons	Lucide React

Module status — what is built

Module	Status	Key features
Weight tracker	✓ Phase 1	Adjusted weight + confidence H/M/L, Tanita BIA flags, 30-day trend, confounder breakdown
Food log	✓ Phase 2	Meals by type, macros, daily progress vs targets, presets
Sleep	✓ Phase 3	Duration auto-calc, quality 1–5, HRV, 30-day trend
Training log	✓ Phase 3	Set-level logging, volume calc, RPE, weekly trend, top exercises
Body measurements	✓ Phase 4	8 measurements, deltas, per-body-part toggle chart

Dashboard	✓ **Phase 4**	Weekly overview, rule-based insights card, backup button, CSV export

What Cowork delivered (March 2026)

Three features were implemented in response to the initial critique:

- Automated backup — POST /api/backup copies dev.db to prisma/backups/ with timestamp. Auto-purges to last 10 files. 'Back up now' button on dashboard shows filename + size on success.
- Rename + confidence label — 'True weight' renamed to 'Adjusted weight' throughout the UI. calculateConfidence() in retentionModels.ts returns High / Medium / Low based on active confounders. Badge on dashboard card and weight history table.
- Cross-module insights — Dashboard API generates up to 3 weekly observations: scale vs adjusted gap with confounder attribution, sleep quality impact, training day inflammation, calorie context. Fires when sufficient data exists (≥2 weight entries, ≥3 sleep entries).
- Note: 5 pre-existing TypeScript errors remain (Recharts Formatter type in files not touched). Runtime not affected.

Branding strategy

This section synthesises recommendations from the Enterprise GPT review and the broader critique. Decisions are marked clearly.

Positioning statement

Build the private app that tells serious people what their body weight actually means — especially when travel, training, and real life make the scale noisy.

Feature hierarchy:

1. Core promise: know what you actually weigh
2. Hero feature: True Weight™ / Adjusted Weight
3. High-salience use case: Trip Mode
4. Trust layer: private, local-first, no cloud
5. Supporting modules: training, sleep, eating, measurements

What privacy actually is — important reframe

Privacy is a trust multiplier, not the main wedge. Most users do not wake up wanting 'local-first software.' They want relief from a specific pain: the scale keeps lying to me, especially after travel, restaurant food, alcohol, carbs, and hard training. Lead with the pain and the solution. Privacy comes second as the reason to trust the answer.

The name

Decision: keep MY PENS as the internal / developer acronym. The public product should launch under a name from the signal / clarity territory. Options below are positioning examples — trademark clearance required before use.

Option	Name direction	Logic
Recommended	MY PENS (internal) + public brand TBD	Keep MY PENS as the developer/project acronym. Launch under a consumer-facing name from the 'signal/clarity' territory. True Weight™ becomes the hero feature brand.
Alternative A	MY PENS everywhere	Keep the acronym public but lead all messaging with True Weight™ and Trip Mode. The name becomes a curiosity hook, not the value claim.
Do not do	True Weight as master brand	Excellent feature brand. Too descriptive and generic to own strongly as a company name. Keep it as the feature.

Name direction options

Direction	Example names	Why it fits
Signal / clarity	Baseline, Driftless, Clearline	Maps to core promise: know what your body weight actually means. Calm, intelligent, slightly technical.
Weight + signal	Weightline, SignalScale, TrueBase	Combines the primary subject (weight) with the differentiating idea (signal vs noise). Easy to explain.

Drift / noise	Driftless, Unnoised, Steadmark	Positions around what the app removes: the noise. Strong for body-comp audience.
Neutral / minimal	Baseline, Marker, Steadmark	Ownable, broad, can grow with the product. Less literal but very clean.

Tone rules

✓ Tone should be	✗ Tone should not be
Calm and intelligent	*Guilty or shame-based*
Private and non-judgmental	*Diet-culture or restrictive*
Slightly technical / precise	*Macho biohacker futurism*
Honest about uncertainty	*Over-claiming 'science-backed'*
Maintenance-minded on travel	*'Don't get fat on holiday'*
Physiology-informed	*Medical / diagnostic*

Homepage copy structure

Element	Content
Headline	Know what you actually weigh.
Subhead	Private weight tracking that separates real body change from travel, restaurant meals, carbs, alcohol, and hard training.
Proof point 1	Adjusted Weight — physiology-informed estimate with confidence rating (High / Medium / Low)
Proof point 2	Trip Mode — context-aware logging for travel days, flights, and restaurant-heavy periods
Proof point 3	Private by design — local-first, no accounts, no cloud, no subscriptions
Proof point 4	Supporting modules — training volume, sleep & HRV, body measurements, food context
CTA	Download / Self-host — your data, your machine, always.

What not to say publicly (yet)

- 'Science-backed' — use 'physiology-informed' until a backtest harness validates the algorithm against known noisy periods.
- 'True weight' as a literal factual claim — it is an adjusted estimate with known gaps. The feature brand True Weight™ is fine; the in-app label should say 'Adjusted Weight Estimate.'
- 'Insulin tracking' — for a non-medical app this is either overstated (you mean glucose) or serious medical territory. Say 'optional glucose context via CGM import' if you go there.
- 'Cortisol tracking' — consumer cortisol sensing is research-grade. Position around 'recovery load' built from HRV, sleep, training, and travel signals instead.

Travel feature — Trip Mode

Travel is the most emotionally vivid proof point for the core feature, not a separate product category. The scale noise created by travel — sodium from restaurant meals, flight-day fluid pooling, alcohol, jet lag, irregular timing — is exactly the problem MY PENS is built to explain.

The killer user experience:

Someone returns from 4 days abroad. Scale is up 1.8 kg.

Trip Mode shows: adjusted weight roughly flat, medium confidence.

Explanation: restaurant sodium + alcohol + flight day.

That is memorable. That is shareable. That is not another diary app.

What Trip Mode is not

- Not a travel health or wellness app — that category is weak; most holidaymakers do not want to log heavily.
- Not calorie tracking on vacation — never sell this as diet compliance on holiday.
- Not a broad travel pivot — travel is one compelling scenario for the core product.

What Trip Mode is

- A context layer that makes the existing weight model more accurate during travel.
- A maintenance-guardrails tool: stay roughly on track, avoid overreacting to scale noise, understand what is transient, recover faster after the trip.
- The product's best storytelling surface and launch hook.

Trip Mode — the right message

Do not say: 'Don't gain on holiday.'

Do say: 'Enjoy the trip. Keep your bearings. Understand the rebound.'

Trip Mode v1 — feature spec

Trip object fields:

- Trip date range (start / end)
- Destination (optional, for timezone)
- Timezone offset from home
- Trip type: leisure / business / other

Daily trip check-in fields:

- Flight / long travel day toggle (auto-adds retention bump)
- Restaurant-heavy day Y/N (sodium confounder — >70% of dietary sodium comes from prepared/restaurant food)
- Alcohol units (existing confounder, already in model)
- Protein-anchored today Y/N
- Steps (optional import or rough estimate)
- Sleep quality (optional — links to existing sleep module)
- Weigh-in (optional, never required — scale access unreliable on travel)

Post-trip re-entry card:

- Automatic card shown when trip end date passes

- Scale delta vs adjusted delta — side by side
- Confidence level with explanation
- Top 3 contributing factors listed
- 'Your adjusted trend over the trip' mini chart

New confounders added by Trip Mode

Confounder	What it captures
Restaurant / sodium day	Prepared and restaurant food accounts for >70% of dietary sodium. Single restaurant-heavy day can cause 0.5–1.5 kg scale retention. Currently missing from model.
Flight / sedentary day	Long periods of sitting cause fluid pooling in lower limbs. Flag flight days with an automatic retention adjustment.
Timezone shift / jet lag	Disrupts sleep quality, cortisol rhythm, and weigh-in consistency. Captured via trip timezone offset.
Weigh-in time inconsistency	Variable weigh-in times inflate noise. Trip Mode should note when weigh-in was not morning-fasted and lower confidence accordingly.
Illness / inflammation	Travel increases pathogen exposure. Inflammation causes water retention. Optional flag.

Expanded roadmap — what still needs to be done

Band	Item	Detail / rationale
✓ Done	Backup endpoint	POST /api/backup + dashboard button. Timestamped files, auto-purge to last 10. Core data durability addressed.
✓ Done	Rename true → adjusted weight	Removed scientific overstatement. More credible, more honest positioning.
✓ Done	Confidence label (H/M/L)	calculateConfidence() based on active confounders. Badge on dashboard + weight history.
✓ Done	Weekly insights card	Rule-based cross-module synthesis on dashboard. App begins explaining, not just logging.
Quick wins	Fix measurements preset bug	module='training' tag on measurements presets. One-line fix. Affects preset reliability.
Quick wins	Fix 5 pre-existing TS errors	Recharts Formatter type in MeasurementsTrend, SleepTrend, TrainingLog, WeightEntry. Small QA pass.
Quick wins	Calorie/macro targets page	Food log progress bars exist but target-setting UI missing. Low effort, high daily utility.
Quick wins	Body comp trend from Tanita data	Body fat %, muscle mass already stored. Chart view only. Zero new data collection needed.
Quick wins	Add missing confounders to model	Sodium/restaurant meals, flight/sedentary day, illness/inflammation, inconsistent weigh-in timing. Improves adjusted weight accuracy.
Mobile (parallel)	Mobile layout + bottom nav	Build Trip Mode mobile-first from day one — the check-in flow must be thumb-friendly regardless of overall app polish. Full bottom nav and responsive overhaul can follow, but the Trip Mode entry screen itself cannot require a laptop.
True Weight v2	True Weight v2 — explanation layer	Tap any adjusted weight entry and see a breakdown: 'Why did adjusted differ from raw today?' Restaurant sodium, alcohol, hard training, flight day, weigh-in time inconsistency.
True Weight v2	True Weight v2 — confidence band	Show adjusted value + uncertainty range, not a single number. High confidence = narrow band shown. Low confidence = wider band + explanatory note.
True Weight v2	True Weight v2 — backtest harness	Replay historical data against known noisy periods (creatine start, refeed, city trip, deload). Validate algorithm before claiming 'physiology-informed' publicly.
Trip Mode v1	Data model: timestamps + timezones	Date-bucket model breaks on travel. Upgrade to full timestamp + timezone before building Trip Mode. Non-negotiable prerequisite.
Trip Mode v1	Trip object / schema	Trip: date range, destination, timezone offset, trip type (leisure / business). Daily trip entry: flight flag, restaurant-heavy Y/N, alcohol units, protein-anchored Y/N, steps (optional), weigh-in (optional, never required).
Trip Mode v1	Trip Mode UI — entry flow	Minimal daily check-in designed for phone. 4–6 taps max. No calorie logging. No macro obsession. Maintenance guardrails only.
Trip Mode v1	Trip Mode UI — post-trip re-entry	Automatic re-entry card when trip ends. Shows: scale delta, adjusted delta, confidence, top contributing factors.

		The killer experience: 'up 1.8kg, adjusted flat, mostly sodium + alcohol + flight day.'
Trip Mode v1	Trip Mode — sodium confounder	Restaurant/prepared food is >70% of dietary sodium intake. Flag restaurant-heavy days explicitly in the retention model. This is the most impactful missing confounder for travel.
Trip Mode v1	Trip Mode — flight day confounder	Long sedentary periods cause fluid pooling in lower limbs. Flag flight days with automatic retention bump estimate.
Launch prep	Demo dataset — seed a trip	Seed a realistic demo: 4-day city trip with restaurant meals, alcohol, and flight day. Scale +1.8kg, adjusted flat. This is the product's best story. Screenshot it.
Launch prep	Landing page / README rewrite	Lead with: Know what you actually weigh. Not with five modules. Feature hierarchy: Adjusted Weight → Trip Mode → Privacy → Supporting modules.
Launch prep	60-second walkthrough video	Record one trip: log in, log food/alcohol/flight context, see adjusted weight explanation. No narration needed if the UI is clear.
Launch prep	Launch post draft	r/QuantifiedSelf, r/naturalbodybuilding, Product Hunt. Hook: 'Why your scale is always wrong after a trip — and how to fix it.'
Later	Wearable import (steps, HRV, sleep)	Garmin / Oura / WHOOP APIs. Reduces manual logging. HRV essential for recovery layer. After trip mode is stable.
Later	Recovery / stress load layer	Combine HRV, sleep quality, resting HR, alcohol, training load, travel days. More honest than 'cortisol', more actionable than raw numbers. Requires wearables.
Later	Per-exercise history + PR tracker	Progression chart, personal best, volume over time per movement. High retention for lifters.
Later	Training programme builder	Define Push/Pull/Legs etc. 'Start session' pre-fills from template.
Later	CGM glucose import	OTC CGMs (Dexcom Stelo, Abbott Lingo). Import for travel meal response + sleep pattern analysis. Not 'insulin tracking.' Narrow audience, real value for them.
Later	Optional cloud sync tier	Obsidian model: free local core, paid sync/backup/mobile PWA. ~€4–6/mo or €49/yr. Only after product has proven demand.
DB (now)	WAL mode pragma	Add PRAGMA journal_mode=WAL to Prisma datasource or app init. One line. Reduces corruption risk on unclean shutdown, improves concurrent read performance.
DB (now)	Auto-backup on app start	Copy dev.db to backups/dev_YYYYMMDD.db before server starts. Supplement the existing manual backup endpoint. Prisma middleware or Next.js startup hook.
DB (sync phase)	Turso (libSQL) migration	Same SQLite dialect, Prisma adapter exists, free tier is generous. Best path when multi-device sync is needed. Near-zero migration — no schema rewrite.
DB (hosted only)	Postgres (Neon/Supabase)	Only if building a hosted/cloud product. Correct choice at that stage but adds server dependency now. Do not migrate pre-demand. Prisma migration is straightforward when the time comes.
Not building	Direct insulin sensing	No FDA predicate device for wearable insulin analysis. Pre-commercial. Wrong use of effort.

Not building	Direct cortisol sensing	Consumer wearables use HRV proxies, not hormone measurement. Build recovery load layer instead.
Not building	Barcode / food database	Graveyard for motivated personal users. Presets cover 80% of the use case. Not worth the complexity.
Not building	Multi-user / gym plans	Dilution at this stage. Single motivated user is the audience.

Recommended sprint plan

This is a realistic 2-day sprint structure assuming ~8 hours per day. Output is a coherent demo story, not feature sprawl. Goal: a stranger sees the product in 30 seconds and instantly understands why it exists.

When	Focus	Output
Saturday AM	**Positioning + copy**	Final one-line positioning sentence. Homepage headline + subhead. Feature hierarchy decision. Tone rules. Whether MY PENS stays internal-only or goes public as-is.
Saturday PM	**True Weight v2 spec**	Define: all confounder inputs, adjustment rules, missing confounders, confidence logic, explanation layer, test scenarios.
Saturday eve	**Trip Mode design**	Trip schema, required daily fields, minimal check-in flow, post-trip re-entry card design. Mobile nav shell.
Sunday AM	**Build visible parts**	Responsive bottom nav. Trip Mode date range + travel flags. Adjusted weight explanation card. Confidence badge upgrade.
Sunday PM	**Proof package**	Seed demo trip dataset. Take screenshots. Record short walkthrough. Rewrite README. Draft launch post.
Sunday eve	**Ticket backlog**	Convert everything else (PWA, backups, timezones, algorithm validation, packaging) into tracked issues.

Multi-model AI workflow

Recommended split across available models. Do not use all five as equal co-founders — that creates reconciliation overhead. Use one as conductor, specialists for specific domains, one as critic.

Model	Role	What to give it
ChatGPT / GPT-5.4 Pro	Conductor	Product decisions, synthesis, final copy, PRDs, structured implementation plans. Give it: positioning memo, landing page copy, Trip Mode PRD, final synthesis after other models disagree.
ChatGPT Enterprise	Project hub	Source-of-truth for all docs, drafts, task lists. Not a separate intelligence — the shared workspace. Projects group chats and files.

Claude	Repo engineer	Long-context codebase audit, data model review, migration safety, component refactor. Give it: repo + 'find edge cases in Trip Mode and weight logic.'
Gemini	Travel UX	Timezone-aware Trip Mode flows, travel user journeys, Maps/location thinking, long-context UX docs, onboarding alternatives.
Grok	Aggressive critic	'Destroy this positioning.' 'What would fitness Twitter mock?' 'What sounds pseudo-scientific?' Use as contrarian, not PM.

Three-pass rule for each major decision:

- Pass 1: ChatGPT writes the first structured answer.
- Pass 2: Claude or Gemini specialises.
- Pass 3: Grok attacks it.
- Then ChatGPT synthesises the final version. That is enough.

Market context and honest success ratings

Path	Rating	What it requires
Beloved indie / open-source tool	**7–8/10**	GitHub release + Product Hunt + r/QuantifiedSelf. Adjusted weight + Trip Mode could go viral among serious lifters and body-comp people.
Sustainable small business	**5–6/10**	Obsidian model: free local core + paid sync / mobile PWA tier. ~€4–6/mo or €49/yr. Requires proof of demand first.
Mainstream consumer product	**3–4/10**	Most users want phone-first, one-tap logging, auto-sync, huge food database. They tolerate cloud because convenience wins. Not this product's path.

Current limitations (honest)

- No authentication — anyone with machine access can see/edit data. Fine for personal use.
- SQLite only — not designed for simultaneous writers. Fine for personal use.
- Backup implemented but requires manual trigger — auto-backup on app start would be stronger.
- 5 pre-existing TypeScript errors (Recharts Formatter type) in MeasurementsTrend, SleepTrend, TrainingLog, WeightEntry. Runtime not affected.
- Measurements preset bug: module='training' tag — one-line fix pending.
- Date-bucket data model needs timestamps + timezone upgrade before Trip Mode is built.
- No mobile-first layout — bottom nav and larger tap targets needed before travel use is realistic.
- No multi-device sync. Data locked to one machine.
- Adjusted weight model missing: restaurant/sodium confounder, flight day, illness, inconsistent weigh-in timing.